



Building RAG Agents with LLMs

Product Building with Agentic Coding

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본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

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Today, You Will...

Product Building with Agentic Coding

- Create a reusable Skill
- Generate and inspect a page
- Connect Vercel through MCP
- Deploy, verify, and update



Today's Route

Understand → choose → practice → apply

1

Understand

Coding models + their limits

2

Choose

One coding workspace

3

Practice

Skill · MCP · deployment

4

Apply

A URL + one verified update



Building RAG Agents with LLMs

Product Building with Agentic Coding

Code Is Also a Sequence of Tokens

A tiny completion—with a checkable result

```
def add(a, b):  
    return a + b
```

```
add(2, 3)
```

5



Run the example

Producing plausible code and passing the check are different.

Code-Focused Model Families

Examples already in this course: Qwen3-Coder and DeepSeek-Coder-V2

QWEN3-CODER

Qwen3-Coder

updated Dec 31, 2025

Qwen3 Coder WebDev

Generate HTML or React code from your web app description

🔥 ❤️ 1.15k

🌟 Qwen/Qwen3-Coder-480B-A35B-Instruct

📄 Text Generation • 🗄️ 480B • Updated Aug 21, 2025 • ⬇️ 30.3k • ⚡ • ❤️ 1.37k

🌟 Qwen/Qwen3-Coder-480B-A35B-Instruct-FP8

DEEPSEEK-CODER-V2

DeepSeekCoder-V2

updated Nov 27, 2025

🌟 deepseek-ai/DeepSeek-Coder-V2-Instruct

📄 Text Generation • 🗄️ 236B • Updated Aug 21, 2024 • ⬇️ 9.65k • ❤️ 700

🌟 deepseek-ai/DeepSeek-Coder-V2-Base

📄 Text Generation • 🗄️ 236B • Updated Jul 3, 2024 • ⬇️ 2.83k • ❤️ 82

🌟 deepseek-ai/DeepSeek-Coder-V2-Lite-Base

📄 Text Generation • 🗄️ 16B • Updated Jul 3, 2024 • ⬇️ 8.36k • ❤️ 118

Code-Model Ecosystem

Model family overview · not a current leaderboard

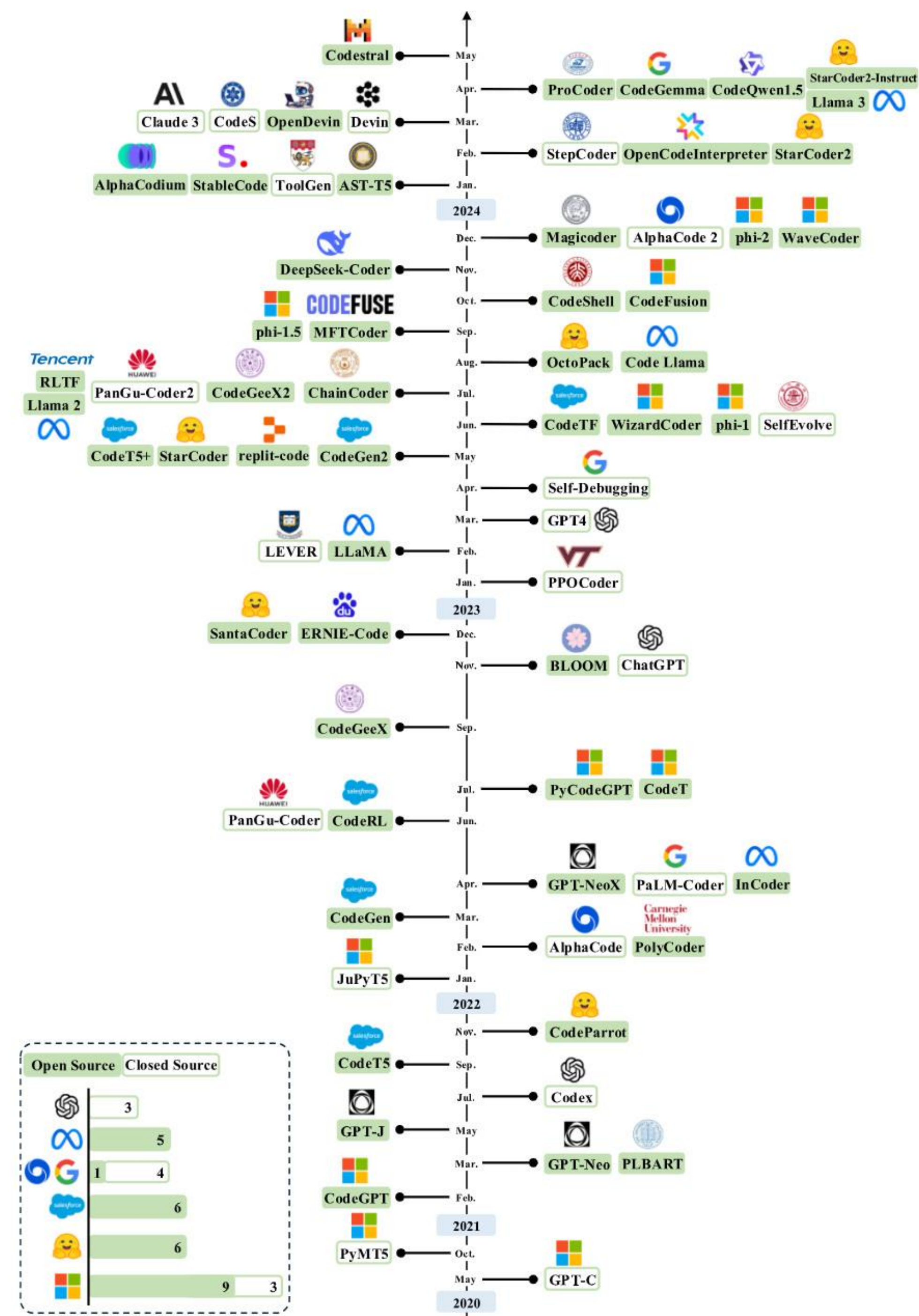
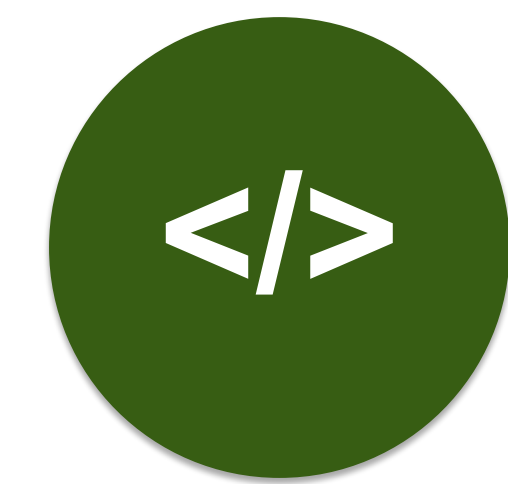


Fig. 1. A chronological overview of LLMs for code generation in recent years. The timeline was established mainly according to the release date. The models with publicly available model checkpoints are highlighted in green color.

Many families.
Different sizes.
Different trade-offs.

Current Models, Concrete Tasks

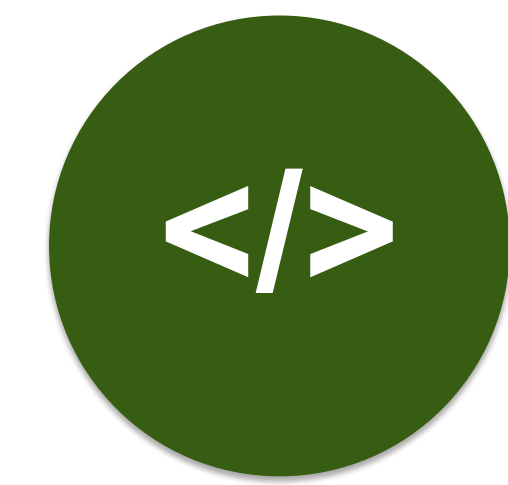
Representative choices · September 2026



GPT-6 Astra



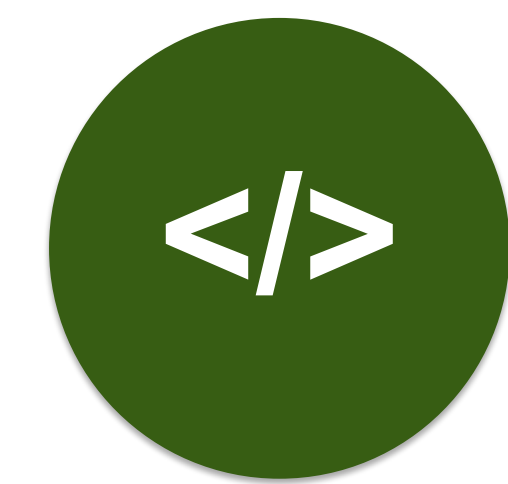
Implement a multi-file feature



Claude Fable 5.1



Trace a bug through a repository



Gemini 3.8 Flash



Iterate on a UI change

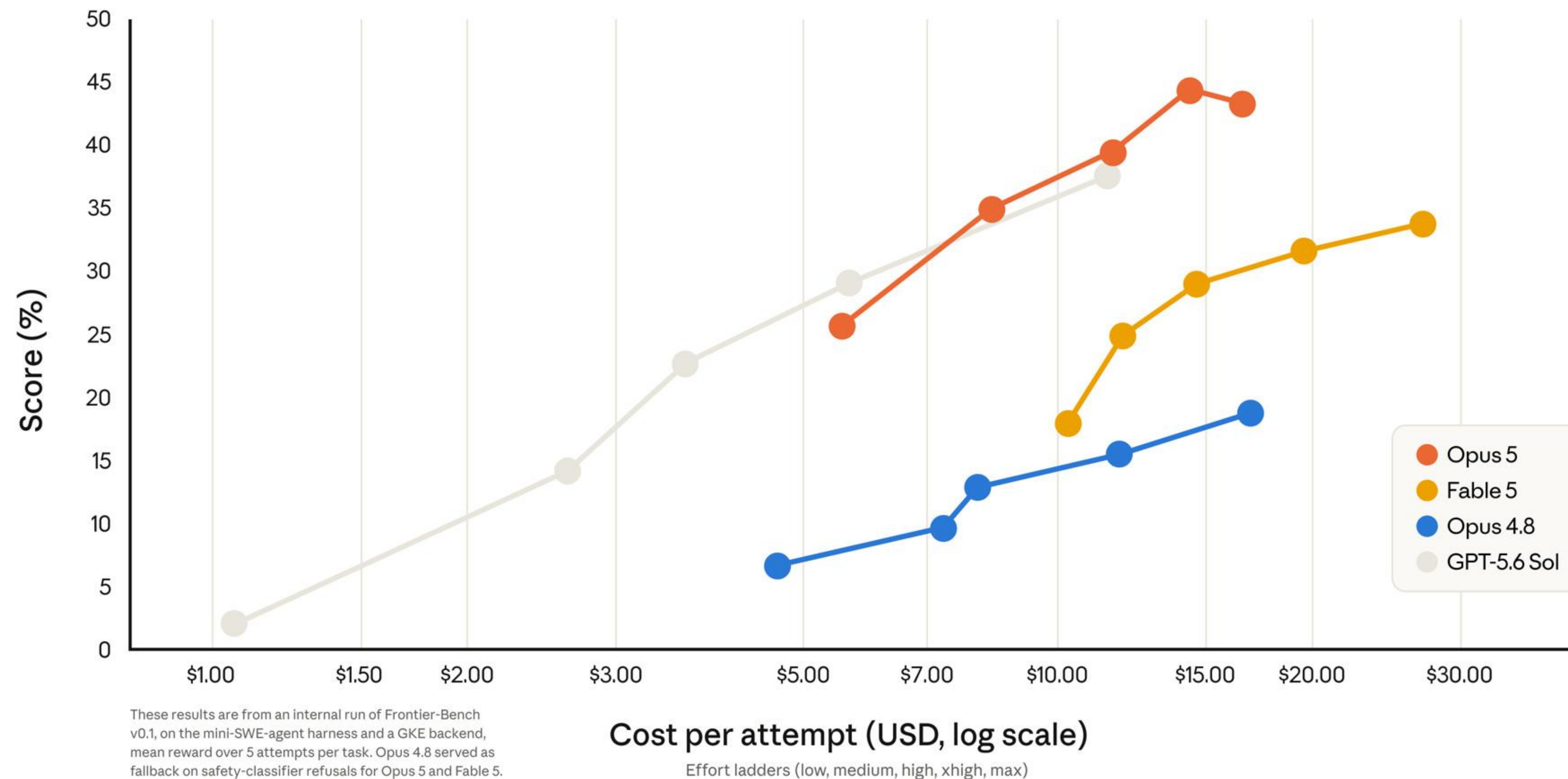
Example tasks—not exclusive capabilities or a ranking.

More Capability Can Cost More

Read both axes: task score and cost per attempt

Agentic coding by effort level

Frontier-Bench v0.1



What Makes It an Agent?

A model inside a tool-using feedback loop



The workflow—not just the model—determines the result.



Building RAG Agents with LLMs

Product Building with Agentic Coding

Vibe Coding vs. Agentic Coding

How you iterate versus how the work is executed

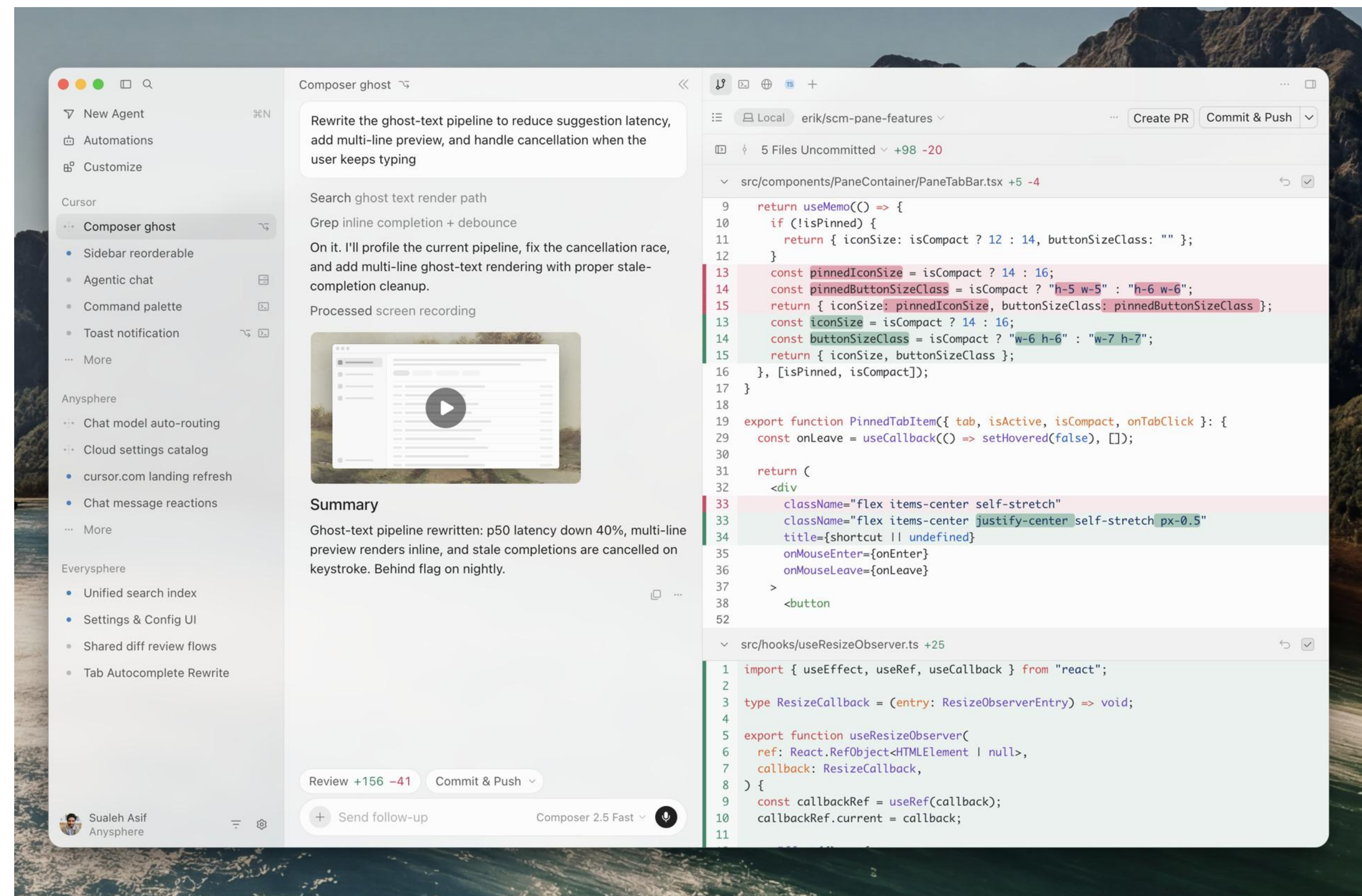
“Make it feel better.”

Intent → visual feedback

Vibe: quick iteration

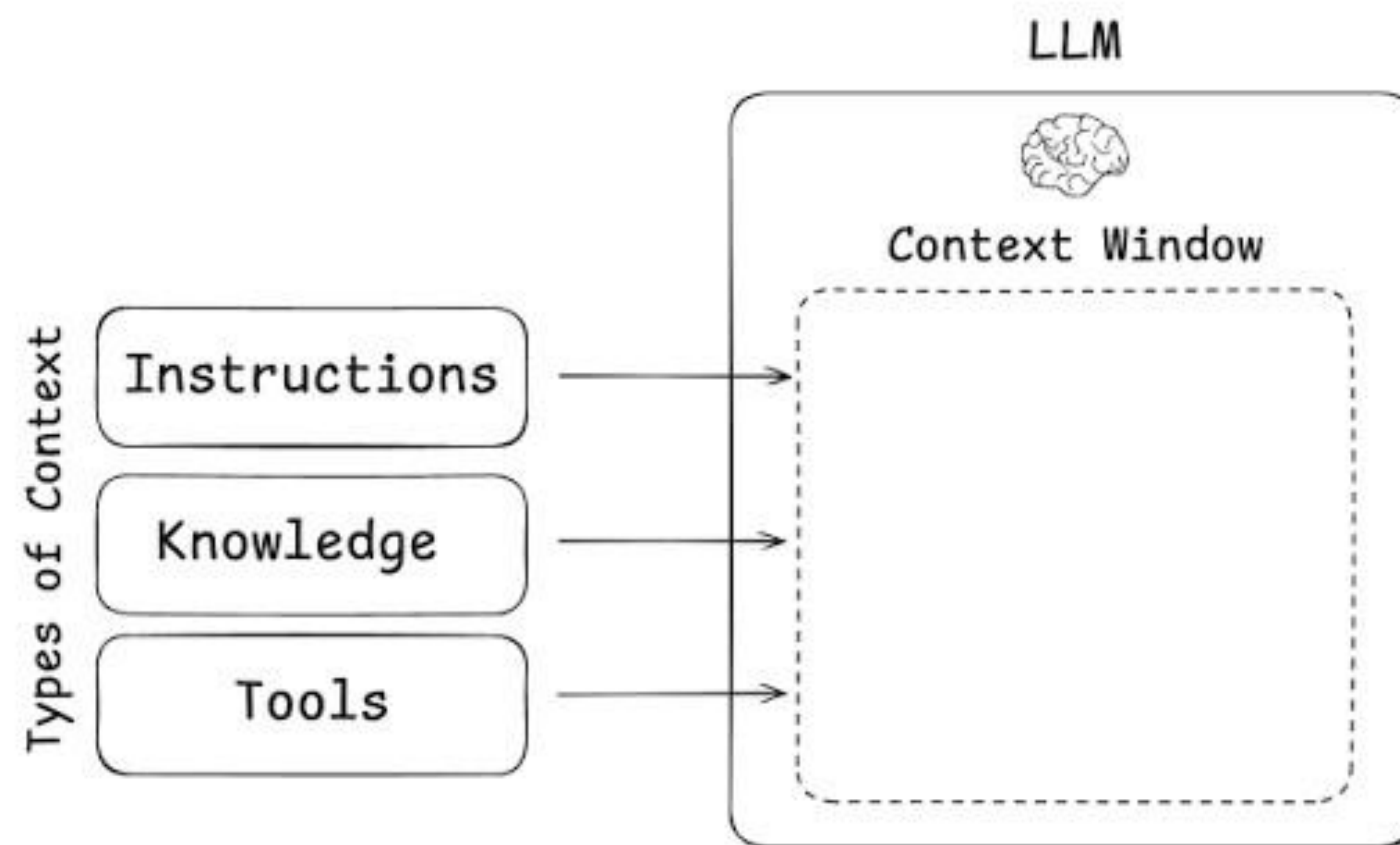
Agent: tool-based work

They can overlap. You still own the acceptance criteria.



Why Vibe Coding Fails

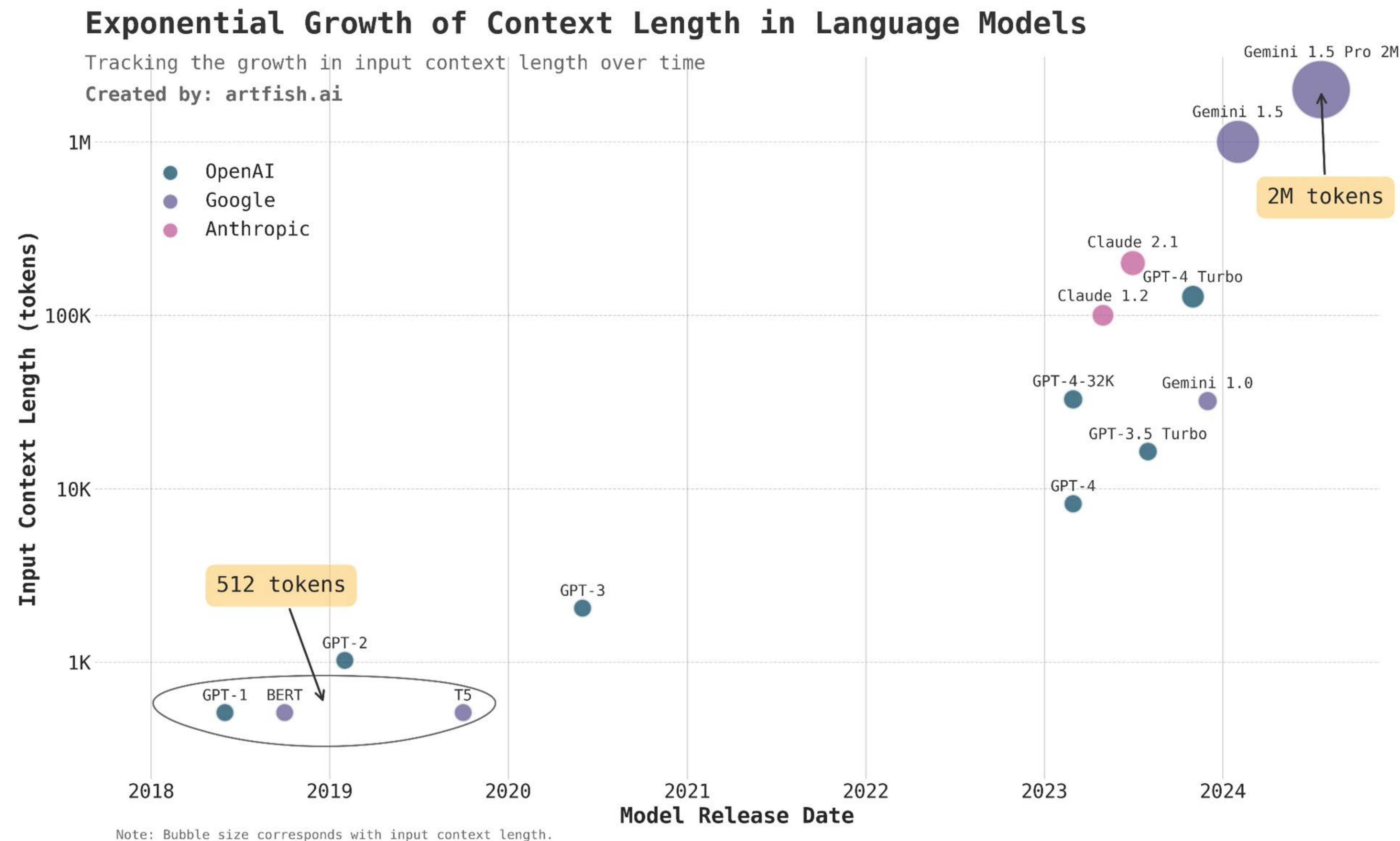
Instructions, context, and verification



- 1 Ambiguous instructions**
- 2 Missing or overloaded context**
- 3 Incorrect model output**

Context Windows Got Much Larger

Historical growth—not a promise of perfect recall



512

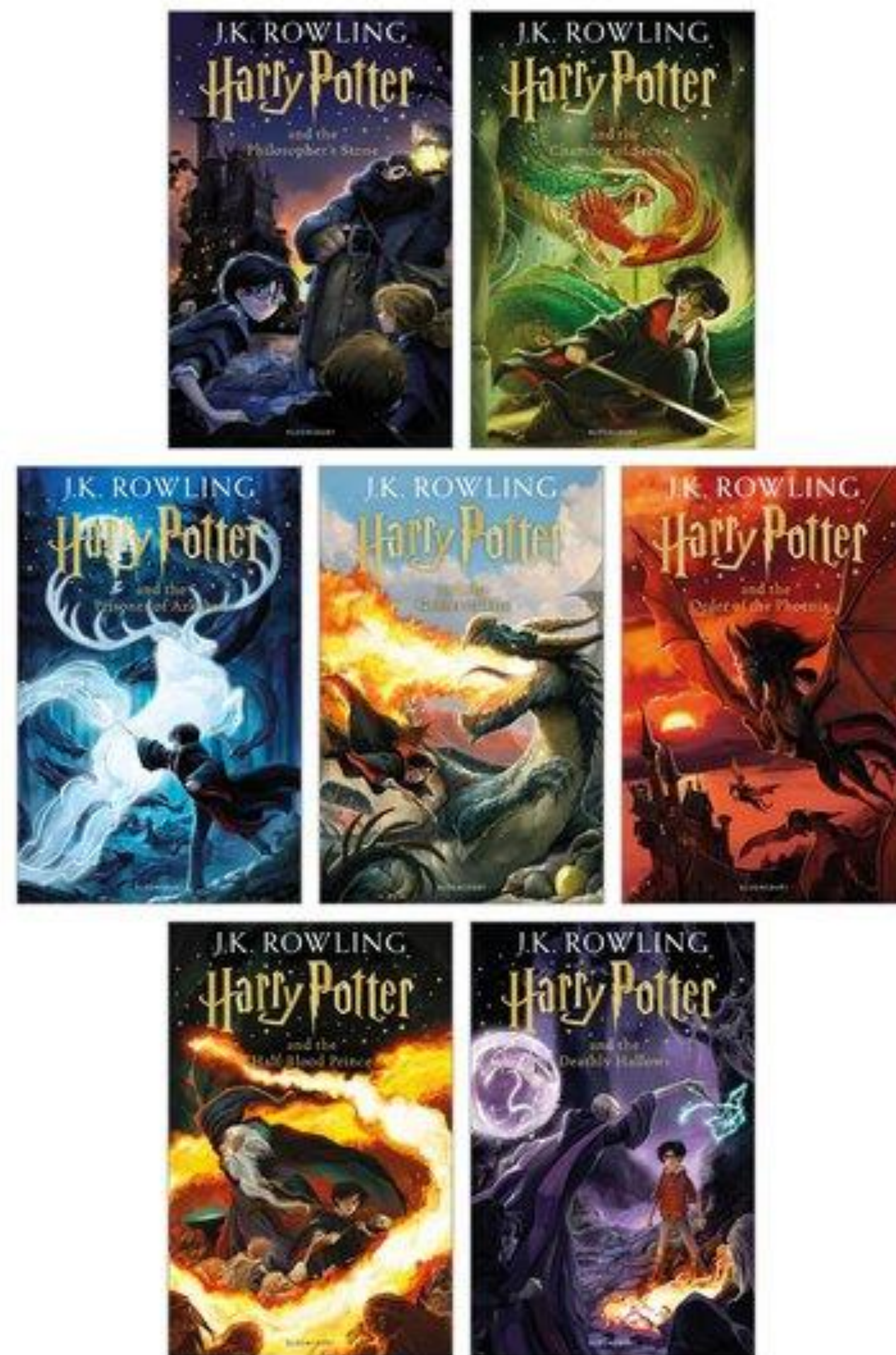


2M

tokens

How Much Is 1.5 Million Tokens?

The Harry Potter example



7 books

~1.5M

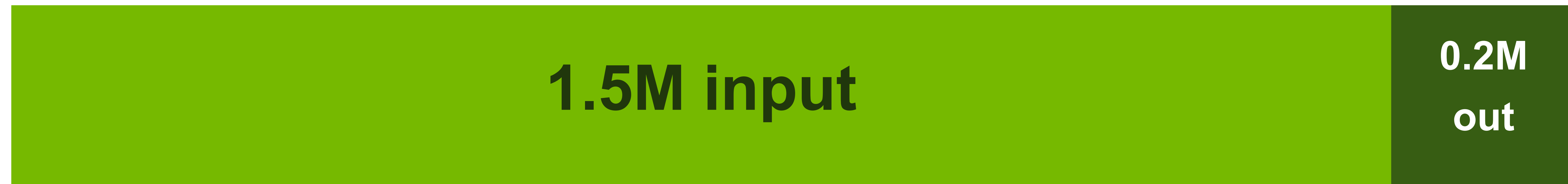
tokens

A scale estimate: token counts depend on edition and tokenizer.

A Large Context Still Has a Limit

Input and generated output must fit the model's budget

Harry Potter thought experiment



Astra context window: 1.05M

This does not fit in one Astra request. Split or retrieve.

Long Context Also Has a Price

An illustrative total across multiple calls



1.5M total input × \$20 / M **\$30**

0.2M total output × \$75 / M **\$15**

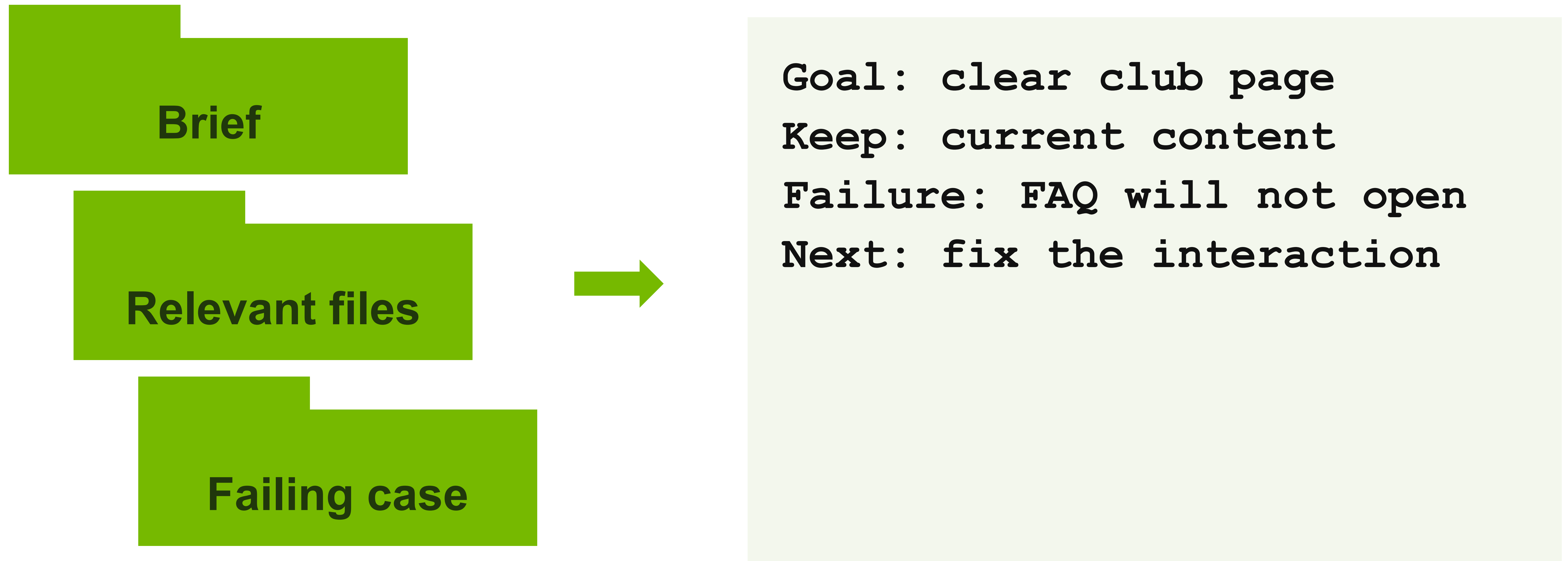
Illustrative total **\$45**

Standard, uncached, long-context rates

Example: 3 calls, each with 0.5M input and ~0.067M output.

Pack Context for the Next Decision

A working folder is more useful than a giant chat history

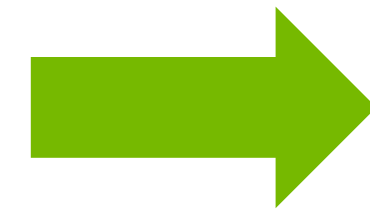


Give the agent the smallest context that explains the next change.

Replace “Make It Better” with a Test

A concrete request changes the conversation

“Improve the app.”

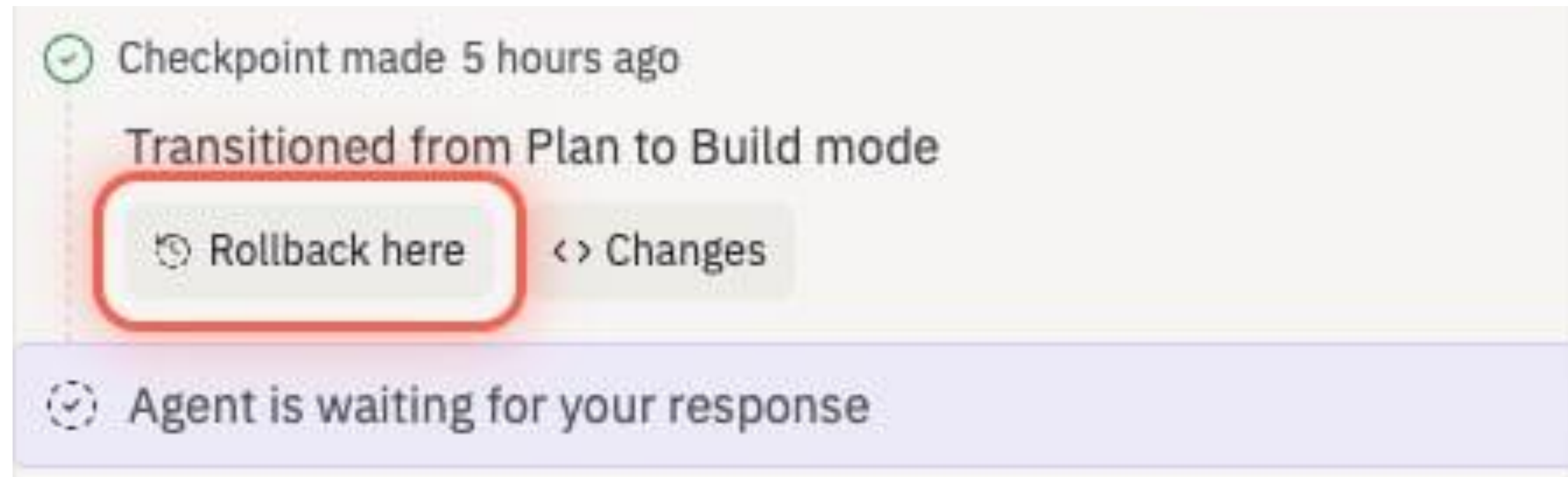


- 1 Open the page
- 2 Click “Explore topics”
- 3 The Topics section appears

An observable outcome beats a long list of adjectives.

Keep a Way Back

Checkpoint before a larger change



Save

Inspect

Restore

Use sample data. Scope permissions. Review before publishing.



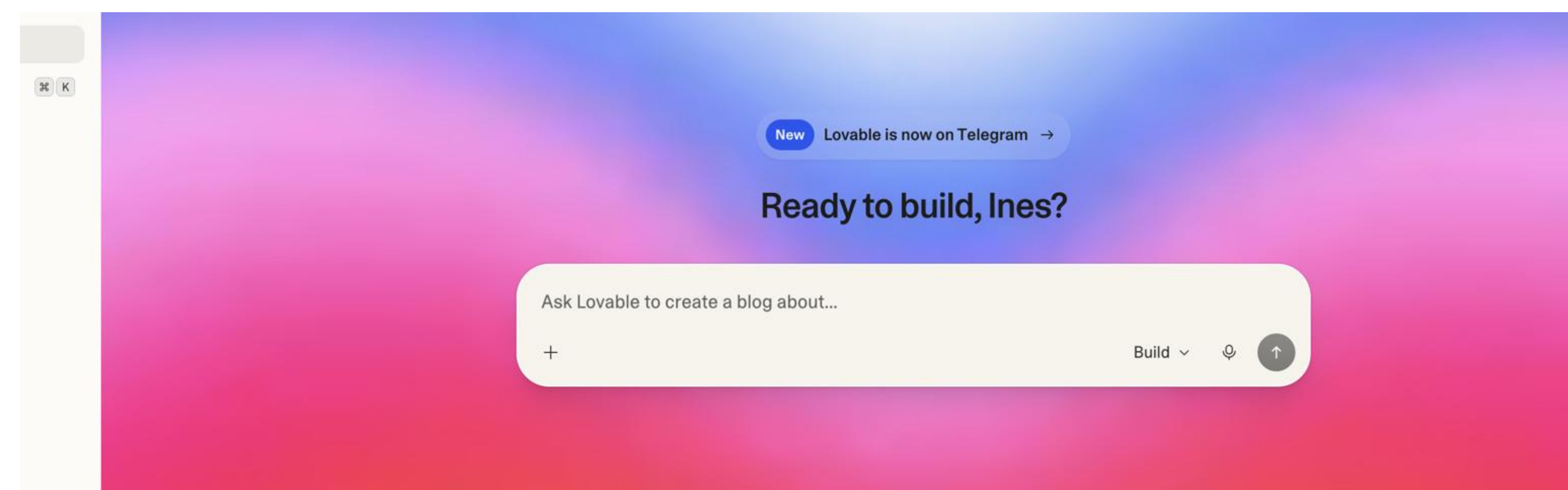
Building RAG Agents with LLMs

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Choose Your Starting Point

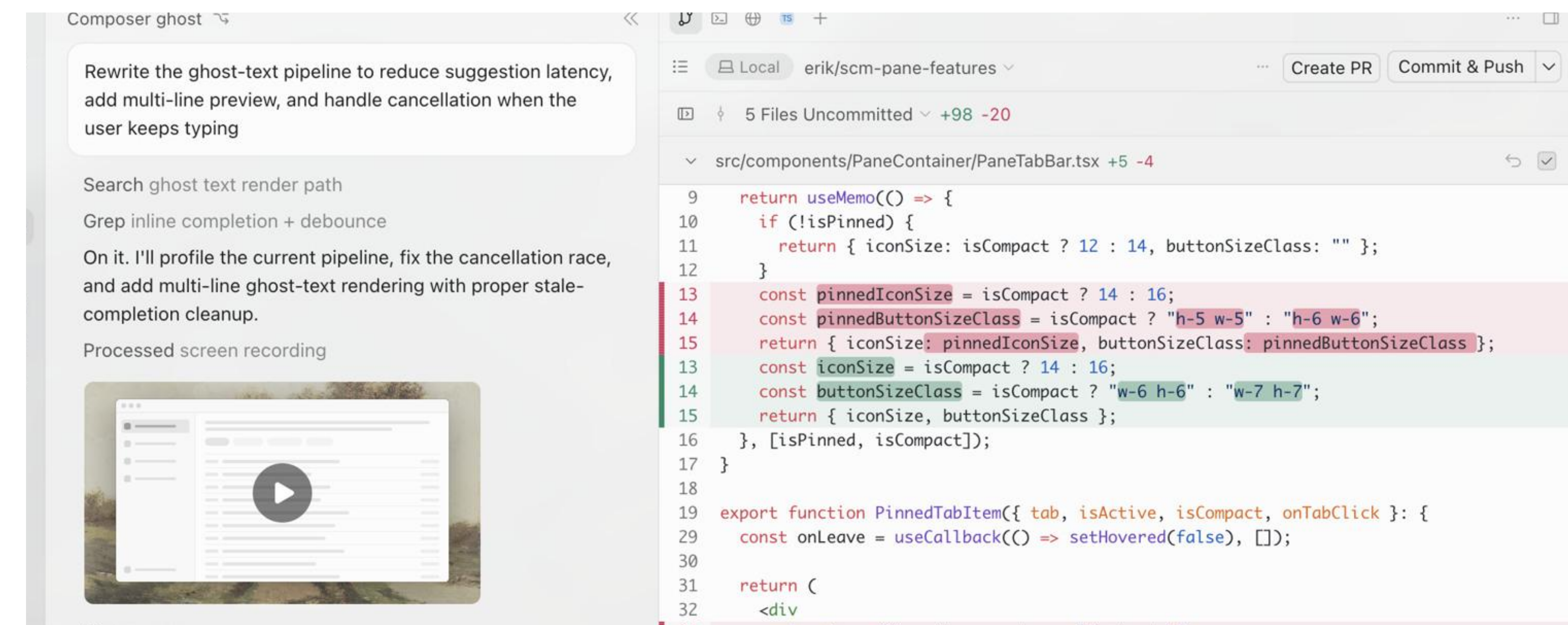
An idea-first builder or a repository-first agent

IDEA → PREVIEW



Lovable · Replit

FILES → PATCH

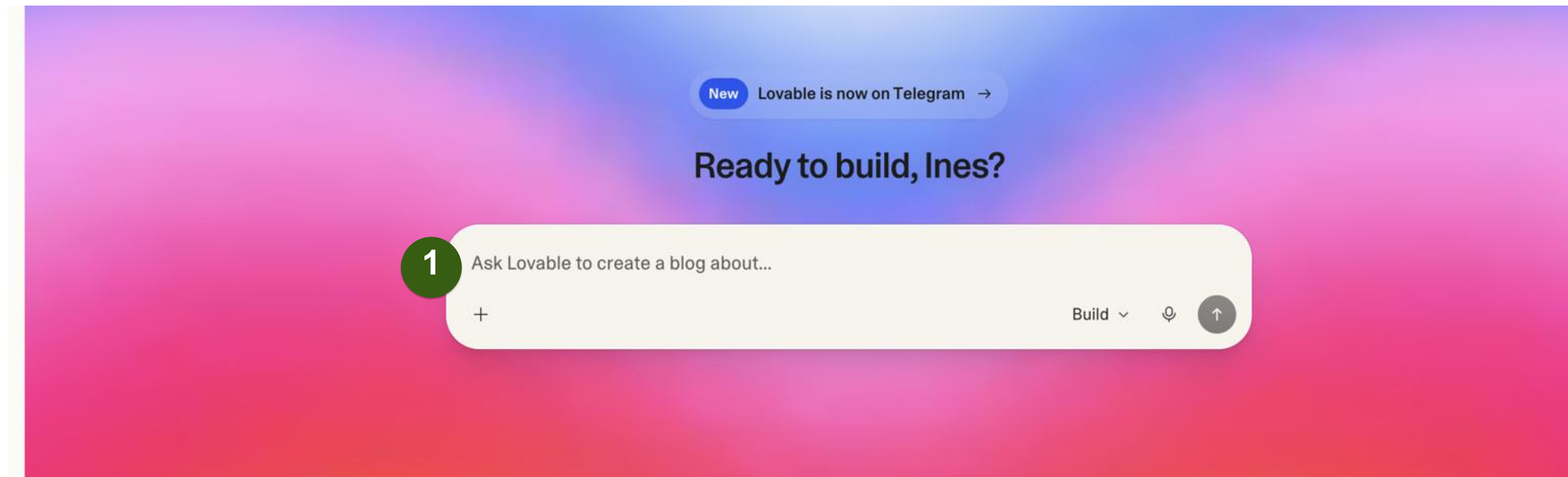


Cursor · Claude Code · Codex

Builders: demo · ChatGPT desktop Work: follow-along

Lovable: Describe the App

Watch where the first instruction goes



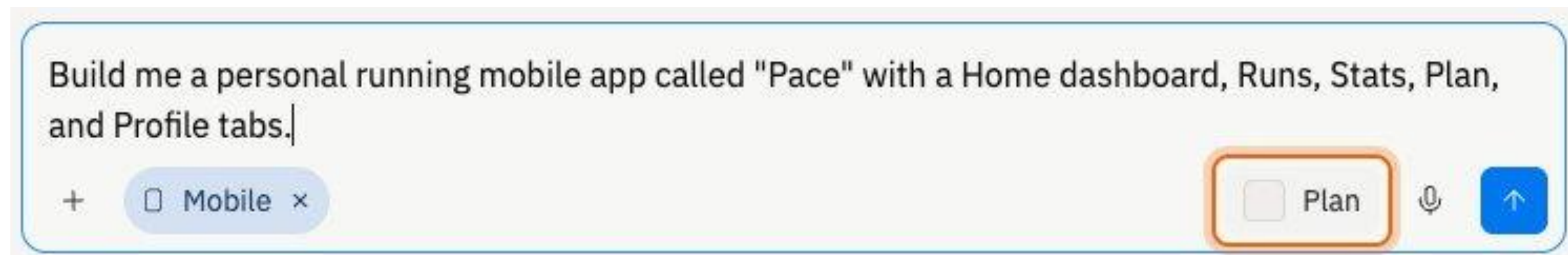
1 Describe the user, outcome, and constraints.

[Open](#) ↗

Replit: Plan Before Build

Read the plan before starting implementation

WRITE THE REQUEST

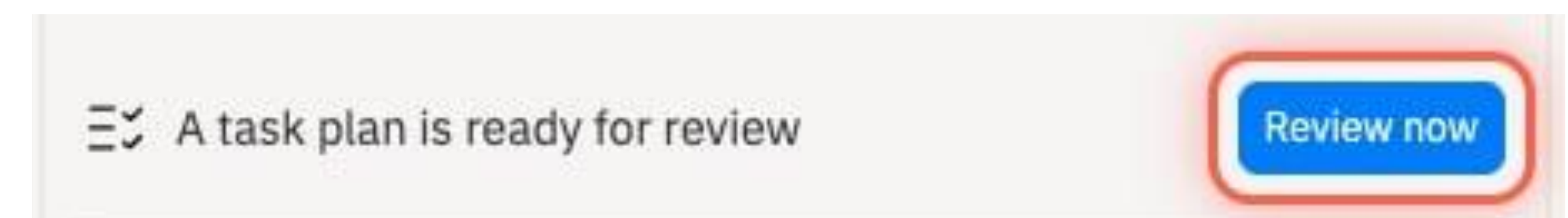


Build me a personal running mobile app called "Pace" with a Home dashboard, Runs, Stats, Plan, and Profile tabs.

+ Mobile x

Plan

REVIEW THE PLAN



A task plan is ready for review

Review now

Turn on Plan

Review now

Check scope, completion criteria, and what stays unchanged.

Cursor: Request → Diff

Check whether the patch matches the request

REQUEST

Rewrite the ghost-text pipeline to reduce suggestion latency, add multi-line preview, and handle cancellation when the user keeps typing

CODE DIFF

```
9   return useMemo(() => {
10     if (!isPinned) {
11       return { iconSize: isCompact ? 12 : 14, buttonSizeClass: "" };
12     }
13     const pinnedIconSize = isCompact ? 14 : 16;
14     const pinnedButtonSizeClass = isCompact ? "h-5 w-5" : "h-6 w-6";
15     return { iconSize: pinnedIconSize, buttonSizeClass: pinnedButtonSizeClass };
13    const iconSize = isCompact ? 14 : 16;
14    const buttonSizeClass = isCompact ? "w-6 h-6" : "w-7 h-7";
15    return { iconSize, buttonSizeClass };
16  }, [isPinned, isCompact]);
```

Inspect changed lines. Ask for a smaller patch if the scope drifts.

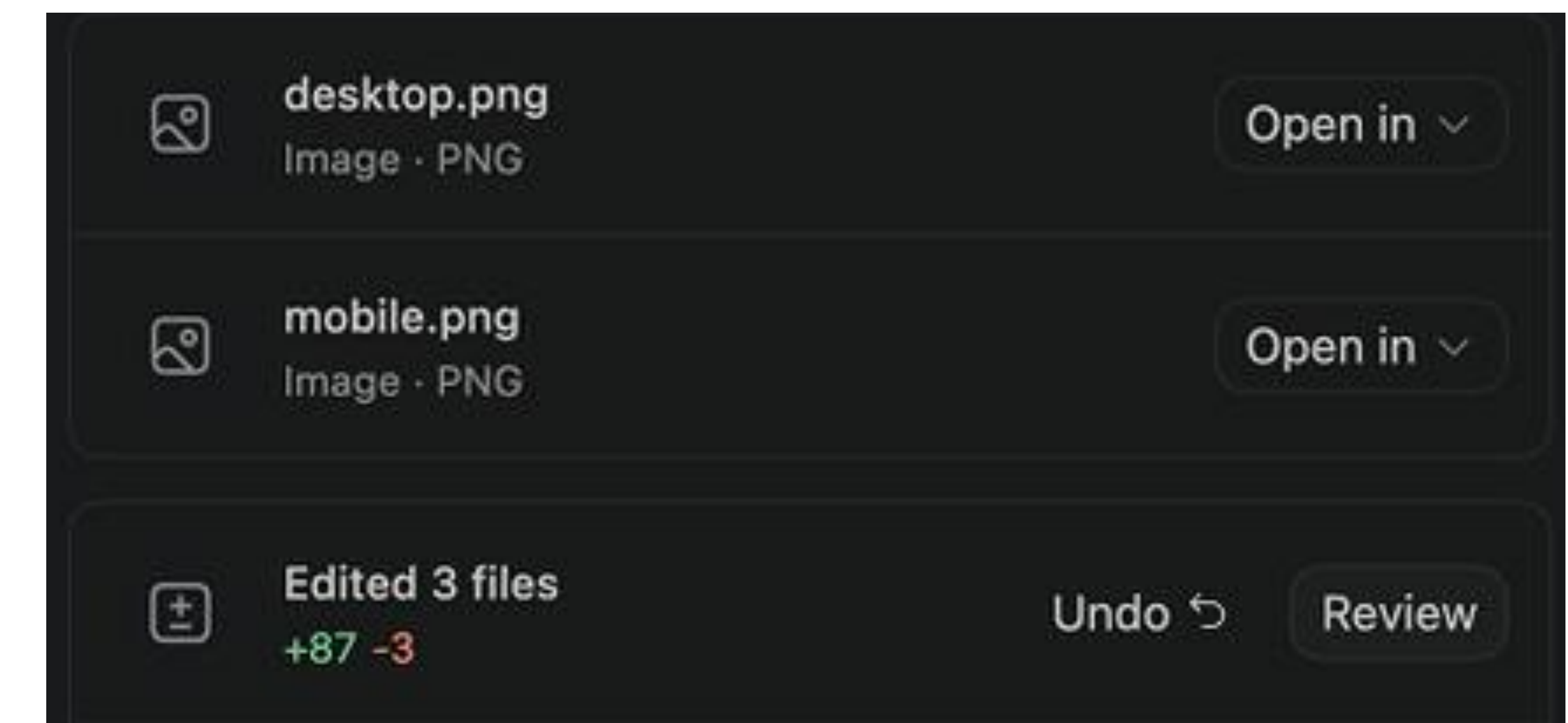
Codex: Review the Change and the Evidence

Look for the diff, result files, and review controls

CHANGED LINES

```
src/components/react/illustrative/CodexSurfaceFeatureIllustration.react.tsx +1 -1 v
441 unmodified lines
442   return (
443     <div className="grid h-full place-items-center p-[6cqw] text-[#161616] dark:text-white">
443     <div className="grid h-full place-items-center p-[8cqw] text-[#161616] dark:text-white">
444     <div className="grid h-full w-full place-items-center rounded-[2.6cqw] border border-black/10 bg-white shadow-[0 16px 45px rgba(0,0,0,0.1)]">
```

RESULTS + REVIEW



Read the actual patch

[Codex ↗](#)

Open the results

[Claude Code: terminal route ↗](#)



From a Skill to a Live Page

Create → connect → deploy → verify

One Workflow, Three Useful Outcomes

A reusable procedure, a working page, and a deployment URL

1 Skill

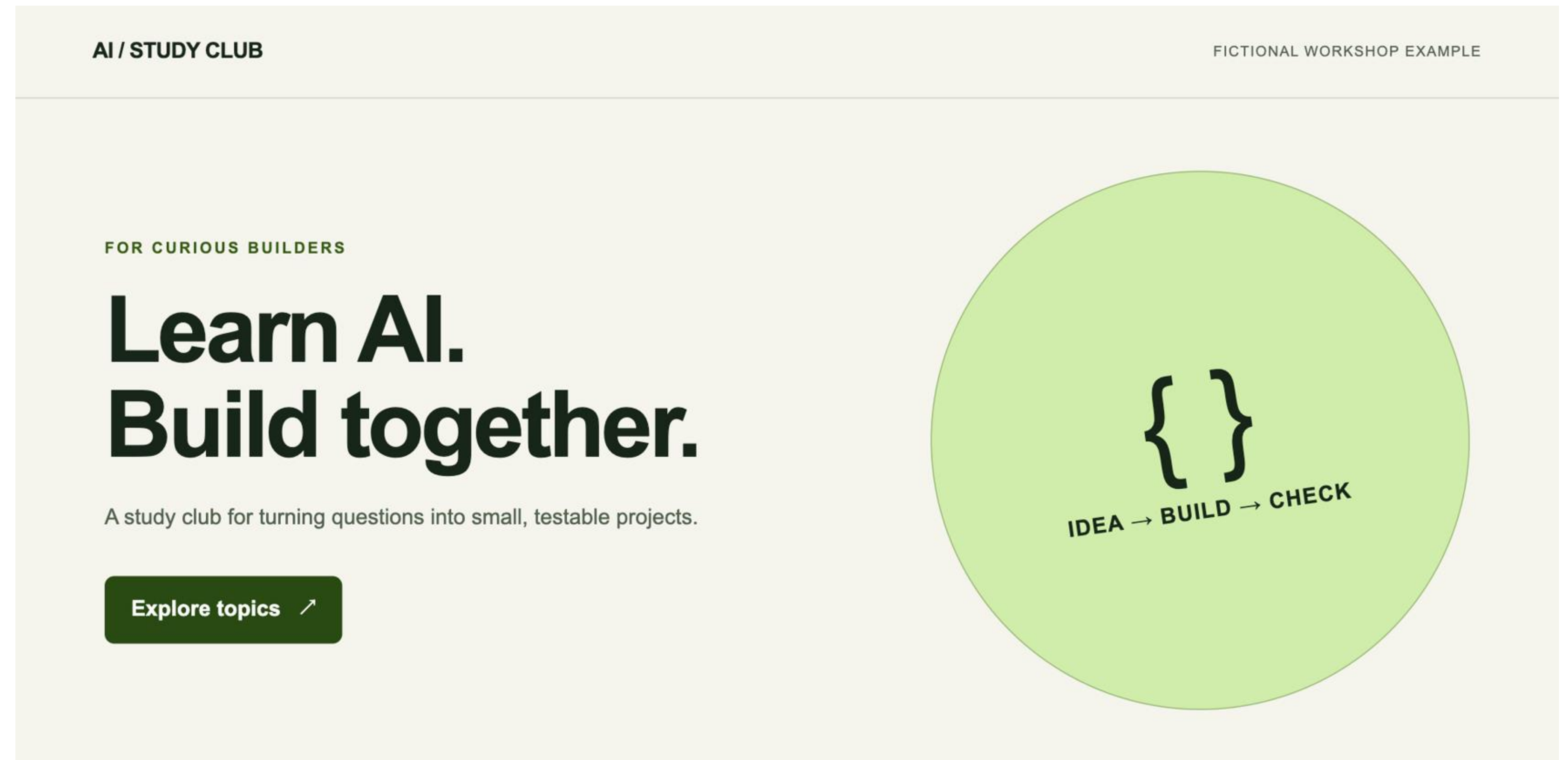
Repeat a useful procedure

2 MCP

Use an external tool

3 Vercel

Publish the reviewed files



Example output · build your own page from EXAMPLES.md.

Start in ChatGPT Desktop Work

One main route for this follow-along

- 1 Open Work in a new task folder
- 2 Check Skills and @skill-creator
- 3 Check Settings → MCP servers

Web uses plugins; local Skills and MCP settings do not sync into it.

Only Three Files in the Practice Pack

Read the brief. Copy a prompt. Check the result.

READ_ME.md

PROMPTS.md

EXAMPLES.md

Setup + recovery

Numbered requests

Skill + inputs A / B

[Download the practice pack ↗](#)

Prompt, Skill, MCP

A request, a procedure, and a tool connection

PROMPT

What I want now

“Make a club page.”

SKILL

How to repeat it

Brief → page →
checks

MCP

**What the agent
can use**

Vercel deployment
tools

Skills: A Reusable Procedure

Define it once; invoke it on different inputs

```
my-skill/
├── SKILL.md           # Required: metadata + instructions
├── scripts/          # Optional: executable code
├── references/        # Optional: documentation
├── assets/            # Optional: templates, resources
└── ...               # Any additional files or directories
```

Instructions + optional supporting files, packaged for reuse.

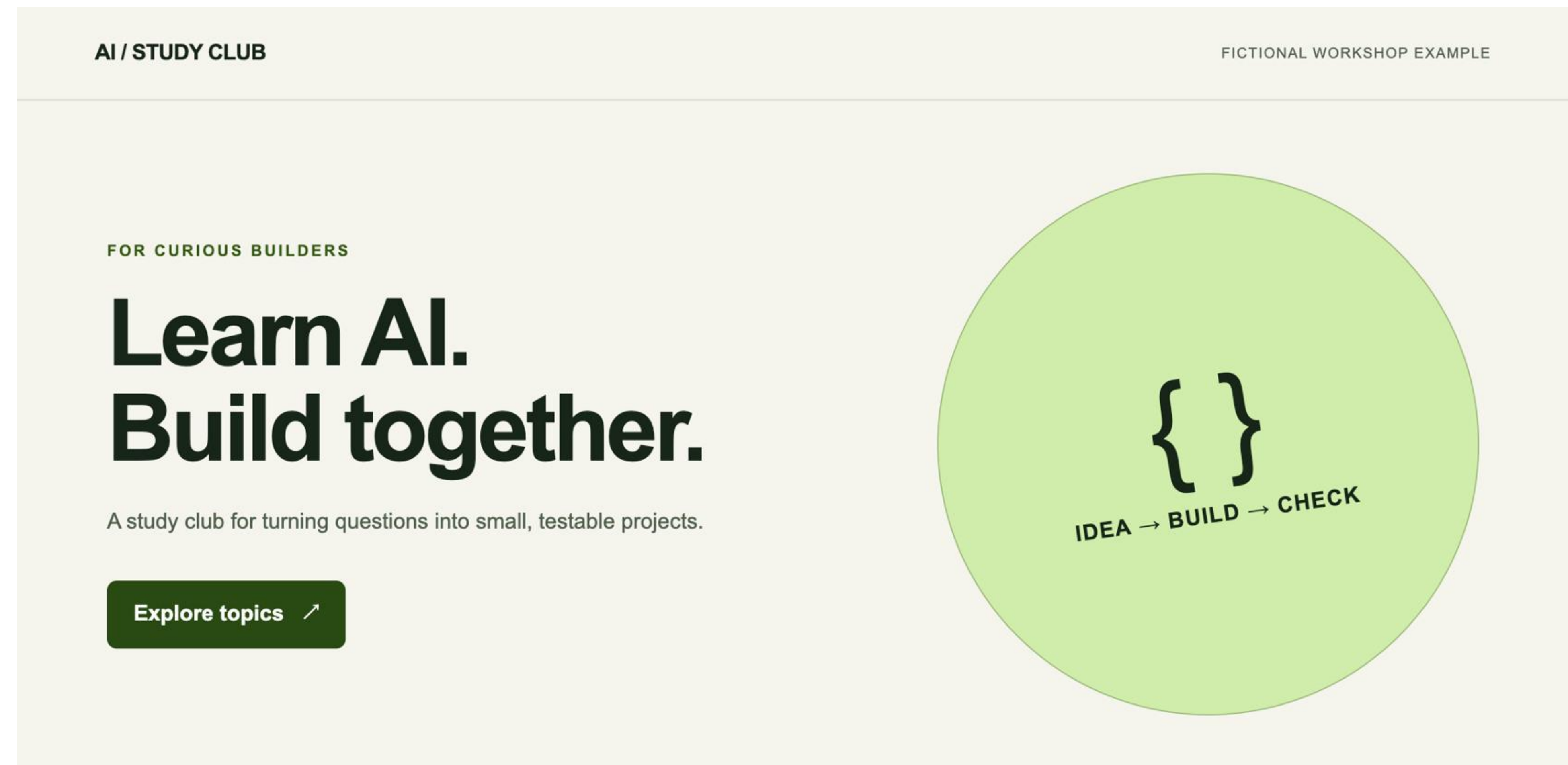
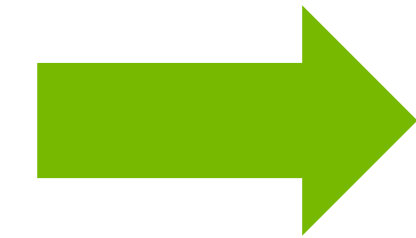
Our Skill: Brief → Page

brief-to-page · useful for clubs, projects, and personal pages

Audience

Facts

Primary action



One static page. No login, database, or invented event details.

Create the Skill

PROMPTS 1 · select @skill-creator

```
Create an instruction-only Skill:  
brief-to-page  
Use the procedure in EXAMPLES.md.
```

Review its trigger, steps, and boundaries.

The example text is not installed until your host creates the Skill.

What Is Inside SKILL.md?

A small procedure—not an application or API connection

```
---
```

```
name: brief-to-page
```

```
description: Turn a public brief  
             into a one-page static site.
```

```
---
```

```
Read → create → check
```

Trigger

When to use it

Procedure

What to do

Boundaries

What not to invent

Deployment stays separate from this Skill.

Invoke It by Name

PROMPTS 2 · select @brief-to-page and provide Input A

```
@brief-to-page
```

```
Create the page from Input A.
```

```
Open and check it. Do not deploy yet.
```

1 Confirm it appears in the Skill picker

Open the generated file and compare it with the brief.

Same Skill, Different Input

PROMPTS 3 · use Input B in a new task folder

INPUT A

INPUT B

AI Study Club

Research Reading Circle

Learn AI.

Read closely.

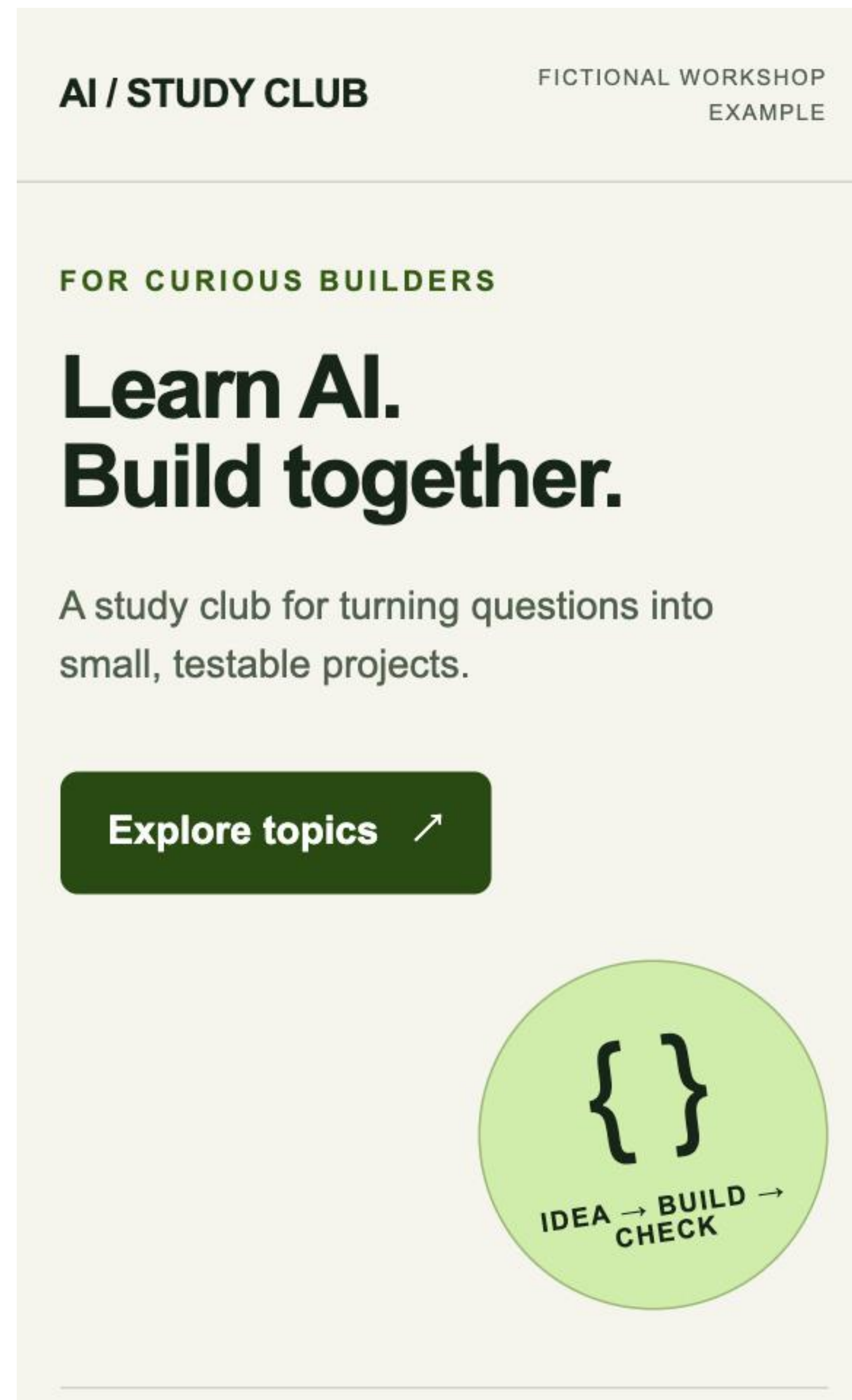
Build together.

Ask better questions.

Keep the procedure. Replace the facts. Do not copy stale content.

Use the Page Before Deploying It

Return to Input A · your layout may differ



1 Explore topics → Topics section

2 FAQ → answer opens

3 Narrow viewport → no clipping

A screenshot cannot prove that a button works.

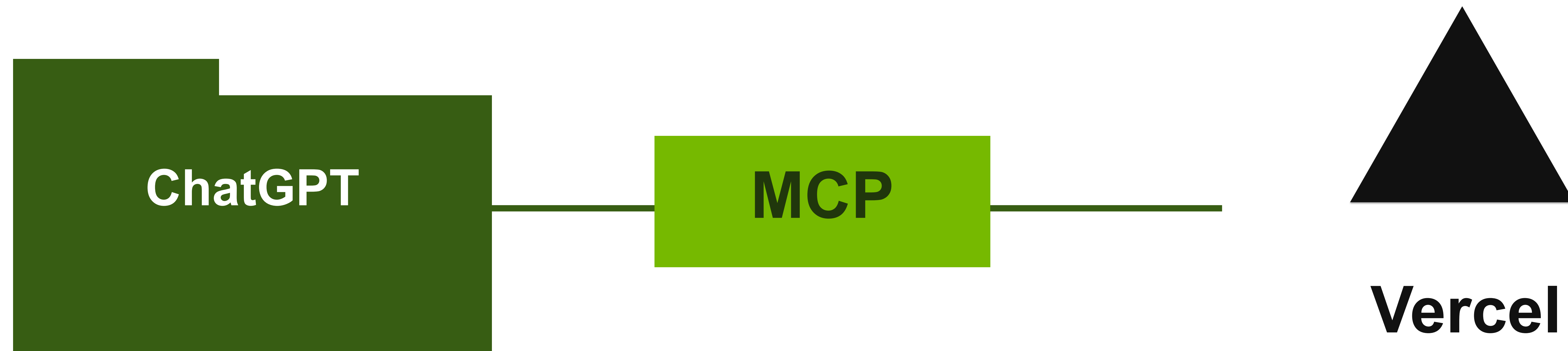


Connect GPT to Vercel

MCP → OAuth → a real tool call

What Does MCP Add?

The model requests an action; the service performs it



Read documentation → inspect projects → deploy files

The Skill tells it how. MCP gives it a tool.

Add the Vercel Server

ChatGPT desktop · Settings → MCP servers → Add server

Name: vercel Type: Streamable HTTP

```
https://mcp.vercel.com
```

1 Save

2 Restart

3 Authenticate

Sign in through the Vercel OAuth window—not a token in chat.

Check the Connection Before Writing

Use /mcp in the desktop composer

- 1 **Confirm vercel is connected**
- 2 **Complete any required OAuth sign-in**
- 3 **Start with a read-only tool request**

Connection check: server status + one successful tool call.

First Tool Call: Find the Deployment Docs

PROMPTS 4 · read-only

```
Use Vercel MCP to find the current docs  
for deploying this static page.  
Explain the required inputs.  
Do not create or modify a project.
```

Evidence: a tool invocation and the returned documentation.

Vercel Can Accept Files Directly

The `deploy_to_vercel` tool receives source files

`deploy_to_vercel`

Deploy files directly to a new Vercel project without a Git repository or the Vercel CLI. Provide the file tree and a deployment target. Vercel creates the project if needed, detects the framework, and starts the build.

FILES

Complete source

PROJECT

Name + team

TARGET

Preview / production

No Git repository or Vercel CLI prerequisite for this route.

Choose a Destination You Recognize

PROMPTS 5 · team first, then a new project name

- 1 List accessible teams
- 2 Select your workshop team
- 3 Choose a unique project name

Use the returned team ID. Do not repurpose an unrelated project.



Deploy, Verify, Update

A real URL—not only generated code

Preview First. Publish Intentionally.

Two deployment targets with different purposes

PREVIEW

Review this version

Main follow-along route

PRODUCTION

Publish the reviewed page

Optional final step

Deployment target and anonymous access are separate checks.

Ask for a Real Preview Deployment

PROMPTS 6 · replace the team and project placeholders

Deploy the reviewed page with Vercel MCP.

Team: <TEAM>

Project: <PROJECT_NAME>

Target: preview

Return the real deployment ID and URL.

Review the destination and files before approving the tool action.

Send the File—not a Description of It

The generated page becomes the deployment input



Complete file contents

At the deployment root

Only the intended source files

Place index.html at the site root.

What Counts as Deployed?

PROMPTS 7 · inspect the deployment record

1

Tool result

Deployment ID + URL

2

Build state

Ready or a real error

3

Live page

Open the returned
URL

Keep the deployment record and open its URL.

Ready Does Not Mean Fully Checked

Build, page retrieval, and user interaction are different

Before you start

▼ Do I need experience?

Basic Python helps; beginners can start with guided examples.

► What will I make?

Open the real URL. Click the link. Expand the FAQ. Resize.

One Change, Then a New Preview

PROMPTS 8 · keep the same team and project

BEFORE

AFTER

Learn AI.

Build together.

Build small.

Learn together.

Review the focused diff. Verify the heading at the new URL.

When the Result Is Not Ready

Use the next piece of evidence

Tool unavailable



Check server + OAuth

Build failed



Read the deployment build logs

**Login page or old
content**



Check access + exact deployment
URL

Keep “generated,” “deployed,” and “verified” separate.

Optional: Publish the Reviewed Version

PROMPTS 9 · make the publication decision explicit

- 1 Use the same reviewed source files**
- 2 Deploy to the intended production target**
- 3 Open the URL with the intended access**

Confirm the final URL and who can access it.

Open-Source Coding Workspaces

The interface is open; model usage may still cost money



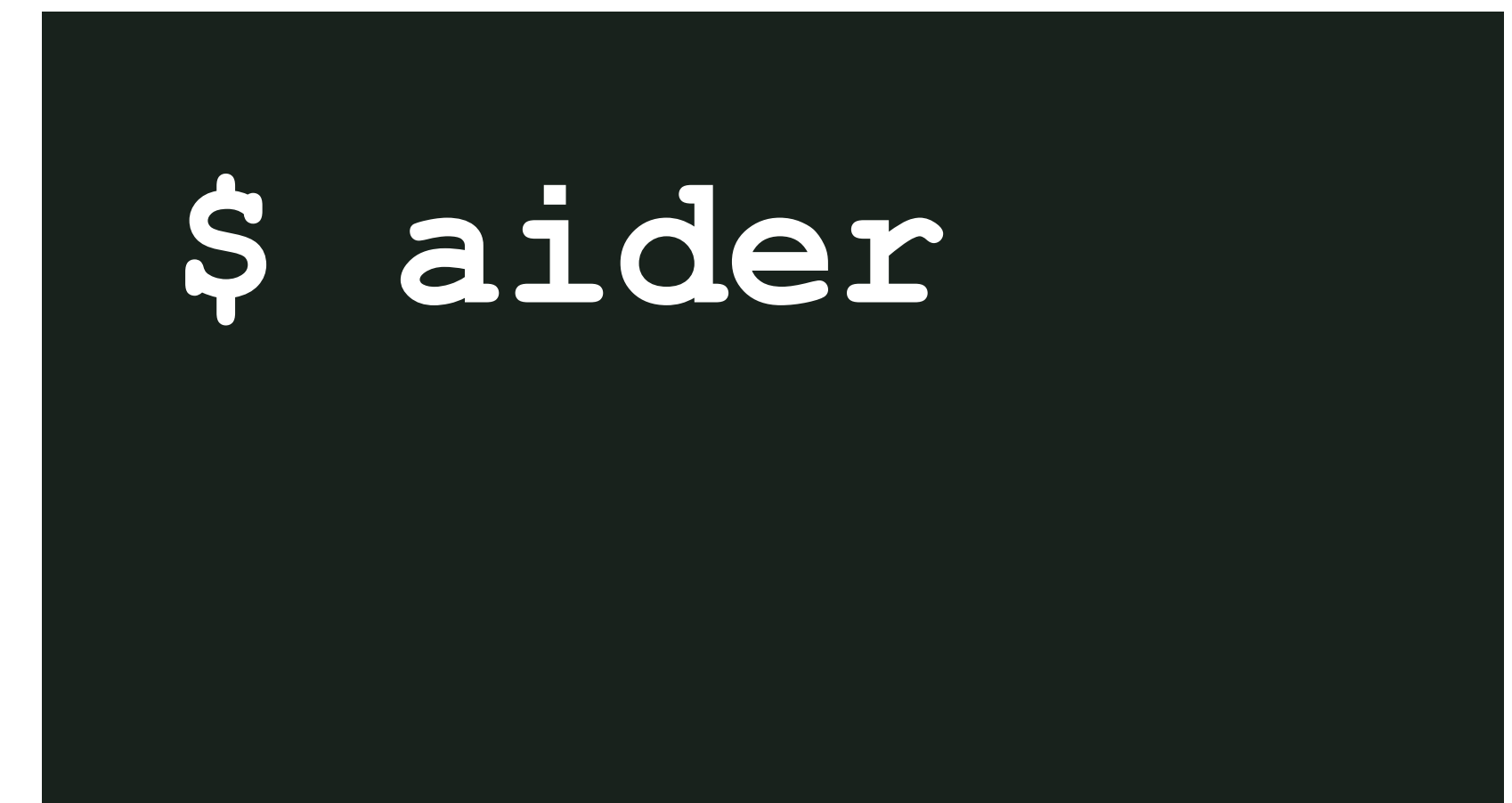
[Cline](#) ↗

Editor + approvals

open|code

[OpenCode](#) ↗

Terminal / IDE / app



[Aider](#) ↗

Terminal + Git

k-skill: Familiar Tasks, Reusable Workflows

Explore a Skill for a task you already understand



Train timetables

HWP documents

Local information

Inspect the procedure, inputs, and required tools. ↗

Spec Kit: Start from a Clear Spec

An optional workflow for a larger project

SPEC



User + behavior

PLAN



Files + steps

TASKS



Small changes

[Spec Kit](#) ↗

Take the Workflow to Your Own Task

The page is one example; the pattern is reusable

SKILL

**Repeat a known
procedure**

MCP

**Connect a useful
service**

EVIDENCE

**Inspect the real
outcome**

Show your Skill, your URL, and one verified update.



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